Parallel Six-Point Compression and a Component–Cycle Span Criterion for Five-Cycle Double Covers

Atharva Vaidya
AI-assisted research draft for independent human verification

July 29, 2026

Resolution status, trust boundary, and AI use. The standard five-cycle double cover conjecture remains open. This manuscript proves an exact reformulation for loopless cubic multigraphs: after a nowhere-zero $\mathbb{F}_2^3$-flow, a Fano plane, and a direction have been chosen, a parallel six-point compression is soluble exactly when an explicit component defect belongs to a circuit-switch span. Existentially choosing those data is equivalent to FiveCDC; no universal choice is proved. The orientable conjecture is not considered.
OpenAI Codex agents using GPT-5-series models, under Atharva Vaidya's direction, discovered the formulation, derived the proofs, wrote the producer and checking programs, ran the computations, conducted an AI-assisted prior-art comparison, and drafted this manuscript. The checker is structurally independent of the discovery implementation, but it is also AI-authored. No human peer review or priority determination is claimed. A human author must check every argument, rerun or independently reimplement the audits, complete a specialist literature review, and accept scholarly responsibility before submission.

Abstract

Let $W = \mathbb{F}_2^3$, let G be a finite connected loopless cubic multigraph, and fix a nowhere-zero W-flow f. A companion six-point one-hole construction asks for a coordinate-pair labelling supported on six points of W, avoiding one additional pair which is then merged. We give a purely combinatorial characterization for the 21 translation classes in which the omitted pair and merge pair are parallel.

Fix a two-plane $H \leq W$ and $0 \neq d \in H$. The edges with values in H form components Q of degree one or three; the complementary edges form circuits on their leaves. A cut defect $\beta_H \in \mathbb{F}_2^{\mathcal{Q}}$ is determined by any one outside value. Each outside circuit C supplies a switch vector γ_C^d, recording the components in which C has an odd number of leaves whose inside-edge value differs from d. We prove that the parallel six-point system is soluble if and only if

$$\beta_H \in \operatorname{span}_{\mathbb{F}_2}\{\gamma_C^d : C\}.$$

These columns are literal flow moves. Adding d to every edge of a chosen outside circuit preserves a nowhere-zero flow and leaves its H-edge subgraph fixed; it changes β_H by exactly γ_C^d. Thus the parallel system is soluble exactly when a set of whole-circuit d-switches reaches the Hušek–Šámal five-point condition $\beta_H = 0$. Equivalently, on every outside circuit one chooses one of two d-paired colour classes so that the chosen edges become Eulerian after the H-components are contracted.

We also prove that existentially choosing f, H, d in this criterion is equivalent to the existence of a standard five-cycle double cover. We explain precisely how this differs from Hušek–Šámal, Theorem 3.16: their five-point condition is the special case $\beta_H = 0$, whereas the circuit columns can cancel a nonzero defect. The graph-level existential statements are nevertheless equivalent.

A standard-library checker independently generates all 900 GL$(3, 2)$-orbits of nowhere-zero flows on a rigid 12-vertex graph and compares the original affine and reduced decisions on all 18,900 parallel systems, with no disagreement. A separate AI-authored SAT census records an odd 21-flag selector in every feasible star fibre of all 4,461 simple biconnected cubic graphs through order 16. These computations audit and motivate the reformulation; they do not resolve the universal conjecture.

1 Definitions, conventions, and exact scope

A *cycle* in this paper is an Eulerian edge-subset, not necessarily connected or 2-regular. A labelled five-cycle double cover is an indexed family $(C_1, \dots, C_5)$ of cycles such that every edge lies in exactly two members. Empty members are allowed, so "five" means "at most five." This agrees with the formulation of Hušek–Šámal [1, Definition 1.2]. We work with finite connected loopless cubic multigraphs. Parallel edges are distinct; loops are excluded. General-graph reductions and the orientable variant are outside the present statement.

Put $W = \mathbb{F}_2^3$, written additively. A map

$$f : E(G) \longrightarrow W - \{0\}$$

is a nowhere-zero W-flow when

$$\sum_{e \ni v} f(e) = 0 \qquad (v \in V(G)).$$

At a cubic vertex the three incident values are distinct and form the nonzero part of a unique two-plane $L_v \leq W$.

The companion preprint [2] proves the following potential description. A compatible pair labelling assigns

$$P_e \in \binom{W}{2}, \qquad \sum_{x \in P_e} x = f(e),$$

such that each coordinate has even local degree. For an incidence $e \ni v$, put $p = f(e)$ and choose either of the other two incident flow values $s_{v,e}$. All compatible labellings are parameterized by potentials $t_v \in W$, with

$$P_{v,e} = t_v + s_{v,e} + \langle p \rangle, \tag{1.1}$$

subject to

$$t_u + t_v + s_{u,e} + s_{v,e} \in \langle f(e) \rangle \qquad (e = uv). \tag{1.2}$$

The choice of $s_{v,e}$ does not affect the pair.

Choose disjoint pairs $R, M \in \binom{W}{2}$. The six-point one-hole system additionally requires every local triangle to avoid the points of R and every edge label to differ from M. Identifying the two members of the unused pair M then gives five coordinate names [2]. Translation of all eight points preserves the problem.

We study the *parallel* classes. After translation they have the unique normal form

$$R = \langle d \rangle = \{0, d\}, \qquad M = a + \langle d \rangle = \{a, a + d\}, \qquad H = \langle a, d \rangle, \tag{1.3}$$

where $H \leq W$ is a two-plane and $0 \neq d \in H$. There are seven planes and three nonzero directions in each, hence 21 parallel flags (d, H).

Remark 1.1 (provisional novelty statement). Hušek–Šámal already prove an exact flow formulation of FiveCDCand the component criterion discussed in Section 6. The companion preprint already proves the general six-point affine system and its parallel/skew local tables. The possible additional contribution here is narrower: the component–circuit elimination of the *parallel* six-point system, the exact whole-circuit switch normal form relating it back to Theorem 3.16, and the elementary proof that the parallel subclass alone remains graph-level exact. We have not found this precise factorization in the cited sources, but our search is incomplete and we make no claim of priority.

2 The component and circuit data

Fix f, H, d as above and set

$$E_H = \{e : f(e) \in H\}, \qquad F = E(G) - E_H. \tag{2.1}$$

At a vertex v, either $L_v = H$, in which case its E_H-degree is three, or $L_v \cap H$ is a line, in which case its E_H-degree is one. Call the latter vertices *leaves*. The graph F has degree two at every leaf and degree zero elsewhere, so its nonempty components are circuits. Write $\mathcal{Q}$ for the components of $(V(G), E_H)$ and $\mathcal{C}$ for the F-circuits.

For a leaf v, let p_v be the value of its unique incident E_H-edge. For $Q \in \mathcal{Q}$ and $C \in \mathcal{C}$, define

$$\gamma_C^d(Q) = |\{v \in V(Q) \cap V(C) : p_v \neq d\}| \pmod 2. \tag{2.2}$$

Choose any $w \in W - H$, and define

$$\beta_H(Q) = \left|\delta_G(Q) \cap F \cap f^{-1}(w)\right| \pmod 2. \tag{2.3}$$

The next lemma makes this well defined.

Lemma 2.1 (outside-value independence)**.** *For every $Q \in \mathcal{Q}$, the value in* (2.3) *is independent of the choice $w \in W - H$.*

Proof. Every vertex has degree one or three in Q. The handshaking lemma therefore gives $|V(Q)| \equiv 0 \pmod 2$. Since G is cubic,

$$|\delta_G(Q)| \equiv 3|V(Q)| \equiv 0 \pmod 2.$$

Every edge of $\delta_G(Q)$ lies in F, hence has one of the four values outside H. Write $H = \{0, p, q, p+q\}$ and choose $w_0 \notin H$. Let n_0, n_p, n_q, n_{p+q} be the four boundary multiplicities modulo two. Their sum is zero by the preceding even-cardinality equation. Summing the flow equations over the vertices of Q gives

$$n_0 w_0 + n_p(w_0 + p) + n_q(w_0 + q) + n_{p+q}(w_0 + p + q) = 0.$$

Comparing coefficients in the basis (w_0, p, q) shows

$$n_p = n_q = n_{p+q} = n_0.$$

Thus all four outside colour classes have the same boundary parity. □

3 The parallel component criterion

The local lists for the parallel normal form (1.3), proved in the companion paper, are

$$\mathcal{A}_L(R,M)=\begin{cases} W-H, & L=H,\\ M, & d\in L,\ L\neq H,\\ R, & d\notin L.\end{cases} \tag{3.1}$$

For completeness, this table can be checked directly. A local triangle $(t+L)-\{t\}$ must avoid R and must not have M as a side. If $d\in L$, the two omitted points lie in one L-coset; if $d\notin L$, they lie in opposite cosets. Separating $L=H$ from the other cases gives (3.1).

Theorem 3.1 (parallel component criterion). *For a fixed nowhere-zero W-flow f and parallel flag (d,H), the six-point one-hole potential system* (1.1)–(3.1) *is soluble if and only if*

$$\beta_H\in \operatorname{span}_{\mathbb{F}_2}\{\gamma_C^d : C\in\mathcal{C}\}\subseteq \mathbb{F}_2^{\mathcal{Q}}. \tag{3.2}$$

Proof. Choose $b\notin H$. Let v be a leaf, with inside-edge value p_v. By (3.1),

$$y_v := t_v+p_v\in M.$$

Thus there is a unique $z_v\in\mathbb{F}_2$ such that

$$y_v=a+z_vd. \tag{3.3}$$

Consider an F-edge of value q joining leaves u,v. At either endpoint, formula (1.1), using p_u or p_v as the other incident value, writes its side as $y+\langle q\rangle$. The endpoint sides agree exactly when

$$y_u+y_v\in\langle q\rangle.$$

The left side belongs to H, while $q\notin H$, so this holds exactly when $y_u=y_v$. Hence z_v is a single common variable z_C on each circuit $C\in\mathcal{C}$.

We now encode sides on the E_H-edges. An outside pair of difference $p\in H-\{0\}$ is one of the two cosets of $\langle p\rangle$ in $b+H$. Give $b+\langle p\rangle$ port bit one and the other pair port bit zero. At a degree-three E_H-vertex, the four potentials $t\in b+H$ give precisely the four even-parity triples of incident port bits. Equivalently, writing $t=b+h$ and the incident values as $p,q,p+q$, a direct substitution in (1.1) shows that the three port bits have xor zero.

At a leaf v, write either incident outside value as $q=b+h$. The port bit on the p_v-edge equals one precisely when

$$a+h+z_Cd\in\langle p_v\rangle. \tag{3.4}$$

At $z_C=0$, condition (3.4) holds exactly when one of the two incident F-edge values is $b+a$. Adding d toggles the condition exactly when $p_v\neq d$. Consequently the forced leaf bit is

$$\ell_v(z_C)=c_v+z_C\mathbf{1}[p_v\neq d], \tag{3.5}$$

where $c_v=1$ precisely when an incident outside edge has value $b+a$.

Within a connected component Q of E_H, common edge port bits satisfying xor zero at every degree-three vertex exist exactly when

$$\sum_{\substack{v\in V(Q)\\ v\ \text{leaf}}}\ell_v(z_{C(v)})=0. \tag{3.6}$$

Necessity follows by summing all vertex equations. For sufficiency, solve edge bits successively along a spanning tree of Q; the only remaining equation is (3.6). A single edge between two leaves has the same equality condition, and a component without leaves has no obstruction.

The coefficient of z_C in (3.6) is $\gamma_C^d(Q)$. Its constant term counts incidences in Q of $(b+a)$-valued F-edges. An edge internal to Q contributes twice and a boundary edge once, so the constant term is $\beta_H(Q)$, by Lemma 2.1. Thus all component equations are the binary system

$$\sum_{C\in\mathcal{C}} \gamma_C^d z_C = \beta_H.$$

It is soluble exactly when (3.2) holds. $\square$

4 The circuit-switch normal form

Contract every component $Q \in \mathcal{Q}$ to one vertex, retaining the resulting loops and parallel edges. For $C \in \mathcal{C}$ and $x \notin H$, let

$$\delta_C^x \in \mathbb{F}_2^{\mathcal{Q}}$$

be the odd-degree boundary vector of the x-valued edges of C in this quotient.

Theorem 4.1 (whole-circuit switch normal form)**.** *For $z = (z_C : C \in \mathcal{C}) \in \mathbb{F}_2^{\mathcal{C}}$, define*

$$f^z(e) = \begin{cases} f(e) + z_C d, & e \in E(C),\ C \in \mathcal{C}, \\ f(e), & e \in E_H. \end{cases} \tag{4.1}$$

Then f^z is a nowhere-zero W-flow, $E_H(f^z) = E_H(f)$, and

$$\beta_H(f^z) = \beta_H(f) + \sum_{C\in\mathcal{C}} z_C \gamma_C^d. \tag{4.2}$$

Consequently the parallel six-point system for (f, d, H) is soluble if and only if a set of whole outside-circuit d-switches produces a flow f^z satisfying the Hušek–Šámal five-point condition $\beta_H(f^z) = 0$.

Proof. Every leaf of E_H is incident with two edges of its unique F-circuit, so a selected circuit adds $d + d = 0$ to the flow sum there. It changes no edge at a degree-three E_H-vertex. Therefore f^z is a flow. A value outside H, when translated by $d \in H$, remains outside H and in particular remains nonzero. Thus f^z is nowhere zero and its H-valued edge set is unchanged.

Fix $w \notin H$. On a selected circuit, adding d swaps the w- and $(w + d)$-valued edge classes. The change of the defect is therefore $\delta_C^w + \delta_C^{w+d}$. At a leaf $v \in C$, its two outside values differ by p_v; they meet the d-pair $\{w, w + d\}$ in odd parity exactly when $p_v \neq d$. After internal edges of each Q cancel twice, this boundary difference is γ_C^d. Summing over selected circuits proves (4.2). Finally,

$$\beta_H(f^z) = 0 \iff \beta_H(f) \in \operatorname{span}\{\gamma_C^d : C \in \mathcal{C}\},$$

so Theorem 3.1 proves the last assertion. $\square$

Proposition 4.2 (circuit-colour dictionary)**.** *For every $C \in \mathcal{C}$, every $d \in H - \{0\}$, and every $w \notin H$,*

$$\gamma_C^d = \delta_C^w + \delta_C^{w+d}, \qquad \beta_H = \sum_{C\in\mathcal{C}} \delta_C^w. \tag{4.3}$$

Consequently, the parallel system for (d, H) is soluble if and only if one can choose independently on each circuit C either the w-valued or the $(w + d)$-valued edges so that the union of all chosen edges is Eulerian in the contracted quotient.

Proof. At a leaf $v \in C$, the two incident outside values differ by p_v. Their intersection with the d-pair $\{w, w+d\}$ has even size if $p_v = d$ and odd size otherwise. Summing this incidence parity over the leaves of C in each Q, with internal edges counted twice, proves the first identity. The second identity is just (2.3) split circuit by circuit.

Now β_H belongs to the span of the γ_C^d exactly when there are $z_C \in \mathbb{F}_2$ with

$$0 = \beta_H + \sum_C z_C \gamma_C^d = \sum_C \big((1+z_C)\delta_C^w + z_C \delta_C^{w+d}\big).$$

This says precisely that the selected union has zero boundary vector. □

5 The parallel normal form is graph-level exact

Theorem 5.1 (exact parallel reformulation)**.** *A finite connected loopless cubic multigraph G has a standard five-cycle double cover if and only if there are a nowhere-zero W-flow f, a two-plane $H \le W$, and $0 \neq d \in H$ satisfying* (3.2)*.*

Proof. A five-cycle double cover is equivalently an edge labelling by two-subsets of five coordinate names such that each name has even local degree. At a cubic vertex the three duads must be the three sides of a triangle on three names: parity makes their symmetric difference empty, and no two can coincide. Conversely such duad labels recover the five Eulerian coordinate edge-sets.

Suppose first that G has such a labelling. Inject the five coordinate names into W, leaving three unused points

$$B = \{u, v, w\}.$$

Give an edge labelled $\{x, y\}$ the value $x+y$. The local triangle condition makes this a nowhere-zero W-flow. Put

$$R = \{u, v\}, \qquad d = u + v, \qquad M = \{w, w+d\}. \tag{5.1}$$

These pairs are parallel and disjoint. Indeed, $w \neq u, v$; $w + d = u$ would imply $w = v$, and $w + d = v$ would imply $w = u$. Every original triangle avoids R, and no original edge label equals M, because u, v, w are unused. Thus the original labelling is a solution of this parallel six-point system. Translating by u puts (5.1) in the normal form (1.3), with

$$H = \langle d, w+u \rangle.$$

Theorem 3.1 gives (3.2).

Conversely, Theorems 3.1 and 4.1 give a whole-circuit switched flow f^z satisfying $\beta_H(f^z) = 0$. By Hušek–Šámal Theorem 3.16 and Observation 3.15, this already gives a five-cycle double cover.

There is also a direct compression proof. A solution of a parallel system gives compatible duad labels on the six points $W - R$, with M unused. Identify the two endpoints of M. No used duad collapses, because M itself is unused. Each unmerged coordinate retains even local degree, and the merged coordinate has degree equal modulo two to the sum of the two old degrees. The quotient labels therefore give five Eulerian edge-subsets covering every edge exactly twice. □

6 Exact relation to Hušek–Šámal

Hušek–Šámal fix a nonzero functional $\mu \in W^*$, an outside value w with $\mu(w) = 1$, and set

$$F_{\mathrm{HS}} = \{e : \mu(f(e)) = 1\}, \qquad M_{\mathrm{HS}} = f^{-1}(w), \qquad Z = \{v : v \text{ meets } M_{\mathrm{HS}}\}.$$

With $H = \ker \mu$, their graph $(V, E \setminus F_{\mathrm{HS}})$ is exactly our (V, E_H). For a component $Q \in \mathcal{Q}$,

$$|Z \cap V(Q)| \equiv |\delta_G(Q) \cap f^{-1}(w)| = \beta_H(Q) \pmod 2,$$

because an internal w-edge contributes two endpoints and a boundary w-edge contributes one.

Their Theorem 3.16 states that their fixed-flow five-point system is soluble exactly when every such component contains an even number of vertices of Z [1, Theorem 3.16]. In our notation, this is

$$\beta_H = 0. \tag{6.1}$$

It is the clean-defect special case of Theorem 3.1: if (6.1) holds, then (3.2) holds for all three nonzero $d \in H$. For a fixed f, H, our six-point system can also be soluble when $\beta_H(f) \neq 0$, because the circuit columns can cancel the defect. Theorem 4.1 gives the exact relationship: this is the closure of the Hušek–Šámal test under legal additions of d on whole outside circuits. The switched flow f^z has the same E_H-components and satisfies $\beta_H(f^z) = 0$. In the coordinate-pair picture, the same freedom comes from allowing a sixth coordinate and forbidding one merge pair rather than requiring three coordinates to be empty.

The distinction must not be overstated. Hušek–Šámal already prove that the existence of some flow and plane satisfying (6.1) is equivalent to FiveCDC[1, Observation 3.15 and Conjecture 3.19]. Theorem 5.1 gives another existentially equivalent normal form, not a stronger graph-level existence theorem. Their Theorem 3.16 also counts the solutions of the fixed-flow five-point system; the present theorem records solubility, not an analogous count.

7 Computational audits

The proofs above are elementary and do not depend on computation. The following two experiments have different trust boundaries and are reported separately.

7.1 Independent fixed-flow regression audit

The standard-library program `verify.py` in the accompanying directory:

1. decodes the rigid 12-vertex cubic graph `K??FEaKR@oE_` and checks connectedness and bridgelessness;
2. independently constructs its binary cycle space and all 900 $\mathrm{GL}(3,2)$-orbits of nowhere-zero W-flows;
3. for each of the 21 flags, constructs the original vertex-potential affine system directly from local triangles and edge equations;
4. asserts that its scalar local affine rows recover the literal allowed potential set exactly;
5. separately constructs the component system of Theorem 3.1; and
6. checks both decisions, Lemma 2.1, (4.3), and the switch update (4.2) on all $900 \cdot 21 = 18{,}900$ systems.

The result is 18,900 agreements and 4,856 soluble systems. This is a finite regression audit, not a proof of the general theorem.

7.2 Star-fibre selector census

For a vertex-star packing P, let f_P be its fundamental-completion flow and let

$$\chi_{d,H}(P) = \mathbf{1}\Big[\beta_H(P) \in \operatorname{span}\{\gamma_C^d(P) : C \in \mathcal{C}(P,H)\}\Big].$$

A possible universal selection target is to choose P with

$$\bigoplus_{\substack{H \le W,\ \dim H = 2 \\ 0 \neq d \in H}} \chi_{d,H}(P) = 1. \tag{7.1}$$

The xor in (7.1) has 21 terms and is $\mathrm{GL}(3,2)$-invariant. It implies that at least one parallel system is soluble. Universal existence of such a packing is open.

An AI-authored lazy-SAT producer exhaustively searched every feasible rooted star fibre in every simple biconnected cubic graph through order 16. The corpus has 4,461 graphs and 50,894 feasible rooted fibres; a packing satisfying (7.1) was retained for every one. The producer uses exactly-one omission colourings, lazy connectivity cuts, literal spanning-tree checks, fundamental completions, and binary affine elimination. This census is frozen separately at

`../output/six-point-parallel-selector-through16/`.

Its manifest and exact command are recorded in the accompanying `README.md`. The producer and the semantic checks are AI-authored, and the exhaustive run does not have an independently checked SAT proof certificate. The result is therefore finite computational evidence, not a theorem for arbitrary order and not independent human verification.

8 Open selection problem

Theorems 3.1 and 5.1 isolate the remaining universal problem without solving it. For each plane H, the defect β_H depends only on the contracted E_H-component quotient, while the three directions $d \in H - \{0\}$ change the circuit columns. One must select a flow—or, in the star-fibre approach, a packing—for which at least one of the 21 defects lies in its corresponding span.

The parity selector (7.1) is deliberately stronger than existence of one soluble flag. Its finite success suggests searching for a local exchange formula under basis exchanges of the three spanning trees. No such formula is proved here. In particular, the fixed-flow characterizations do not eliminate the existential quantifier which carries the original difficulty of FiveCDC.

Reproducibility and disclosure

From this directory, run the fixed-flow checker and rebuild:

```
python3 verify.py --output RESULT.json
SOURCE_DATE_EPOCH=1785283200 tectonic main.tex
shasum -a 256 -c SHA256SUMS
```

The selector package has its own frozen sources, corpus, result, census, and manifest. See `README.md` for its exact check command.

Every proof and program in this package was produced with material assistance from OpenAI Codex agents using GPT-5-series models under human direction. “Independent” refers only to separation between implementations and derivations inside the AI-assisted project. It does not mean independent human verification. The literature comparison, including the interpretation of Hušek–Šámal Theorem 3.16, was also AI-assisted and was checked against arXiv version 1 dated July 27, 2026. Specialist review remains necessary.

References

[1] Radek Hušek and Robert Šámal. Exponentially many circuit double covers, 2026. Version 1, July 27, 2026.

[2] Atharva Vaidya. Six-point one-hole compression for five-cycle double covers. AI-assisted research draft for independent human verification; companion manuscript in the same repository, 2026.